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Module 1 Assignment: Forensic Toxicology

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Lesson A: The Effects of Illegal Drugs upon the Human Body

Use the following information to answer the next question.

Body Parts and Fluids in which Psychoactive Drugs are Found:

1. CNS
2. Liver
3. Urine
4. Blood
5. Kidneys

1. The correct sequence of body parts and fluids, as numbered above, through which psychoactive drugs travel when taken by injection is _____ , _____ , _____ , _____ , and _____. (2 marks)

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Physiological Effects of Psychoactive Drug Abuse:

1. Confusion, insomnia, and impotence
2. Inflammation of heart lining and skin abscesses
3. Disruption of blood flow and increased chance of infection
4. Impairment of one's ability to operate a motor vehicle

2. Match each physiological effect numbered above with the psychoactive drugs given below by placing the number in the effects column. (2 marks)

Psychoactive drug:	Effects:
cocaine	
opiates	
marijuana	
amphetamines	

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3. a. Identify three body parts and/or fluids in which forensic toxicologists are most likely to find psychoactive drugs or their breakdown products? (3 marks)

- b. Psychoactive drugs can also be detected in saliva. Explain why saliva samples are not used regularly by law enforcement officials to detect drug impairment in drivers. (2 marks)

4. Compare and contrast illegal depressants and stimulants in the following ways:

- a. One way that each affects neural activity (2 marks)

- b. The general symptoms caused by each (2 marks)

5. State one possible reason drug-testing devices are not widely used by Canadian law enforcement agencies today. (2 marks)

Lesson B. The Effects of Poisons and Toxins

Use the following information to answer the next question.

Descriptions of Four Groups of Substances

1. harmful compound that is produced in a living organism
2. produced by animals and cause its effect through injection
3. chemical compound causing harmful effects after being absorbed into a body
4. products relief or prevents the effects of a harmful substance

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6. Match the descriptions shown above with the names given below (2 marks)

Substance:	Definition:
poison	
toxin	
venom	
antidote	

Deadly Effects Caused by Various Poisons

1. inhibits ATP production in the brain, spinal cord, and heart
2. prevents oxygen from binding to hemoglobin in red blood cells
3. after intense muscle spasms, stops activity in the medulla oblongata
4. causes multi-organ failure by preventing enzyme and mitochondrial activity

7. Match each description shown above with the poisons given below (2 marks)

Poison:	Effects:
arsenic	
cyanide	
strychnine	
carbon monoxide	

8. a. Describe the harmful effects poison can have on the body. (2 marks)

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b. Identify the two main ways poisons tend to be absorbed into the body. (2 marks)

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9. Explain why poisoning deaths do not always involve the consumption of true poisons. (3 marks)

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10. Discuss why homicides caused by intentional poisonings are rare. (3 marks)

11. a. Why is the actual number of poisonings in North America unknown? (2 marks)

b. How does the cause of reported poisonings compare to the cause of poisonings causing death? (3 marks)

c. What substance is the most frequent cause of reported poisonings? (1 mark)

d. What substance is the most frequent cause of poisonings resulting in death? (1 mark)

Lesson C. Forensic Toxicology Testing Techniques

Use the following list to answer the next question:

Various Steps in Gas Chromatography

- 1. A chromatogram is created using a computer.**
- 2. Isolated sample and gas are injected into tubing.**
- 3. Diffusion of isolated sample and gas into a liquid.**
- 4. Isolated sample is placed in heated injection chamber.**
- 5. Chemical components of isolated sample enter a detector.**

12. The correct sequence of steps involved in gas chromatography, as shown above, are (2 marks)

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Use the following list to answer the next question.

Some Descriptions of Toxicology Testing Techniques

- 1. A chemical reaction may occur to produce a crystalline precipitate.**
- 2. Small test strips are placed in a urine sample and a change in colour occurs.**
- 3. A fragmentation pattern is created from substance's unique chemical composition.**
- 4. Drug-specific antibodies are added to an isolated sample and a reaction is observed.**

13. Match each description shown above with the correct toxicological testing technique given below. (2 marks)

Testing Techniques:

Descriptions:

immunoassay	
microcrystalline	
colour testing	
mass spectrometry	

14. Explain how the testing for drugs or poisons from a living person differ from the testing for drugs or poisons from a dead person. (3 marks)

15. Explain three reasons the initial screening for drugs or poisons must be followed by confirmation testing. Indicate which tests have these problems. (6 marks)

16. Why is mass spectrometry considered the most accurate method of confirmation testing in forensic toxicology? (3 marks)

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Lesson D: Crime Case Studies Involving Forensic Toxicology

Case Study #1: The Poisoning of a Politician

Use the following information to answer the next question.

Various Dioxin Poisoning Symptoms

1. cancer
2. nausea
3. diabetes
4. vomiting
5. heart disease
6. liver damage
7. abdominal pain
8. swollen pancreas

(Record your four-digit answer in descending order.)

17. Some of the long-term effects that Viktor Yushchenko could suffer from because of the dioxin he ingested include _____ , _____ , _____ , and _____ . (2 marks)

18. a. Explain why small amounts of dioxin are found in every human's blood. (2 marks)

- b. State how much dioxin Victor Yushchenko had in his blood stream as compared to the average individual. (1 mark)

Case Study #2: The Jonestown Tragedy

1. coma
2. heart attack
3. convulsions
4. loss of consciousness

(Record your four-digit answer in descending order.)

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19. The correct sequence of events, as numbered above, that each Jonestown poisoning victim would have experienced before death is _____ , _____ , _____ , and _____ . (2 marks)

20. a. Whose visit to Jim Jones' colony in Guyana in November 1978 precipitated the mass suicide/mass murder? (1 mark)

- b. Explain why this person paid a visit to Jim Jones' colony? (2 marks)

Case Study #3: Was it a Drug Overdose or an Intentional Poisoning - That is the Question?

Use the following information to answer the next question.

Various Types of Drugs and Poisons

1. arsenic
2. cocaine
3. marijuana
4. strychnine
5. amobarbital
6. secobarbital
7. Pentobarbital

(Record your four-digit answer in descending order.)

21. The chromatogram of the victim's blood (sample evidence #2) indicates that he had ingested _____ , _____ , and _____ . (2 marks)

22. Did the male victim in this case die due to a drug overdose or due to an intentional poisoning? Support your choice with four pieces of evidence. (4 marks)